

AI Project Specification Sheet

- **Project Name:** [Enter your project name here]
- **Team Name:** [Enter your team name here]
- **Project Duration:** August 15, 2025 (Fri) ~ August 21, 2025 (Thu) (7 days total)
- **Final Presentation Date:** August 22, 2025 (Fri)
- **Project Summary:** [Describe your project in a single, concise sentence. E.g., Developing a deep learning model to detect toxic comments in Korean.]

1. Project Overview

1.1. Topic and Objectives

- **Topic:** [Describe the specific topic of your project. E.g., Korean text sentiment analysis based on Transformer.]
- **Objectives:** [Clearly state the quantitative/qualitative goals you aim to achieve. E.g., Achieve over 85% classification accuracy for 7 sentiment categories (joy, sadness, anger, etc.).]

1.2. System Input/Output Definition

- **Input:** [Describe the form and examples of data input to the model/system. E.g., Korean sentences within 500 characters entered by the user (Text).]
- **Output:** [Describe the form and examples of the final result shown to the user. E.g., Classification results for 7 sentiments and their probabilities for the input sentence (JSON).]

2. Technical Design and Implementation Plan

2.1. Dataset

- **Acquisition Plan:** [Detail how you will obtain the data. E.g., Utilize the 'Korean Emotional Conversation' dataset from AI-Hub, crawl comment data from relevant community sites.]
- **Data Details:** [Describe the type, format, size, and key features of the data. E.g., CSV format, approx. 100,000 sentence-emotion label pairs, composed of 7 emotion classes.]

2.2. Data Preprocessing

[List the steps for processing the collected data for model training.]

1. **Remove missing values and duplicate data**
2. **Remove noise such as special characters and emoticons using regular expressions**
3. **Tokenize using a morphological analyzer (e.g., KoNLPy, Mecab)**

4. **Process stopwords**
5. **Pad and encode to match the model input format**

2.3. AI Model Design

- **Model Selection (Transformer-based):** [Select the model(s) you will use from the list below (V) and briefly state the reason.]
 - [] **BERT** (Bidirectional Encoder Representations from Transformers)
 - [] **GPT** (Generative Pre-trained Transformer)
 - [] **T5** (Text-to-Text Transfer Transformer)
 - [] **ELECTRA** (Efficiently Learning an Encoder that Classifies Token Replacements Accurately)
 - [] **KoBERT / KoGPT / KoELECTRA** (Korean-specific pre-trained models)
 - [] **Other (describe freely):** [E.g., Lightweight models like DistilBERT, ALBERT, or a custom-improved model architecture]
- **Reason for Selection:** [Explain why you chose this model in relation to your project's objectives. E.g., KoBERT is expected to show high performance in classification tasks like sentiment analysis due to its strength in understanding the bidirectional context of Korean sentences.]

2.4. Model Training

- **Training Environment:** [Describe the hardware and software environment for training. E.g., Google Colab Pro (GPU: Tesla P100), TensorFlow 2.x, Hugging Face Transformers library.]
- **Key Hyperparameters:** [List the main hyperparameters to be set. E.g., learning_rate: 5e-5, epochs: 3, batch_size: 32, max_seq_length: 128]
- **Training Strategy:** [Describe the training strategy, such as Fine-tuning, Transfer Learning, etc.]

2.5. Performance Verification

- **Validation Dataset:** [Specify the split ratio for training/validation/test sets. E.g., 80% / 10% / 10%]
- **Evaluation Metrics:** [Specify the metrics to evaluate the model's performance.]
 - **Quantitative Evaluation:** Accuracy, F1-Score, Precision, Recall, Confusion Matrix
 - **Qualitative Evaluation:** Analyze the model's predictions on various real-world examples and document error cases.

2.6. User Interface

- **Implementation Method:** [Describe how users will interact with the AI model's features. E.g., Create a simple web page with a text input box and an 'Analyze' button]

button. Display the results as a bar chart below.]

- **Tech Stack:** [List the frameworks or libraries for UI and server implementation.
E.g., UI - Streamlit, Server - Flask]

3. Team and Schedule Management

3.1. Roles & Responsibilities (R&R)

Name	Role	Key Responsibilities
[Team Member 1]	PM / ML Lead	Project schedule management, lead final presentation, design and lead AI model architecture, training, and validation.
[Team Member 2]	Data Engineer	Dataset acquisition and crawling, data preprocessing and analysis, building data pipelines.
[Team Member 3]	ML Engineer	Write model training and experiment code, hyperparameter tuning, test and analyze model performance.
[Team Member 4]	Backend/UI Developer	Develop API for model serving, design and implement user interface (UI), system integration.

3.2. Project Schedule (Milestones)

Day	Date	Owner	Goals & Deliverables
Day 1	Fri	Entire Team	Finalize project topic and spec sheet, start data acquisition and analysis.
Day 2	Sat	Data Engineer	Complete and share the first version of the data

			preprocessing code.
Day 3	Sun	ML Lead/Engineer	Complete baseline model architecture design and training code.
Day 4	Mon	ML Team	Share results of the first model training, analyze issues, and discuss improvements.
Day 5	Tue	ML Team, UI Dev	Improve model performance (hyperparameter tuning), start UI prototype development.
Day 6	Wed	UI Dev, Entire Team	Integrate model with UI, conduct integration testing, and finalize features.
Day 7	Thu	Entire Team	Final testing, bug fixes, and complete presentation materials.
D-Day	Fri	Entire Team	Final Project Presentation

4. Additional Suggestions and Considerations

4.1. Potential Risks and Mitigation Plans

- **Risk:** [E.g., Model performance may not meet expectations due to insufficient training data.]
- **Mitigation Plan:** [E.g., Apply Data Augmentation techniques or utilize additional similar datasets.]
- **Risk:** [E.g., The model may be biased due to an imbalanced dataset for certain sentiment classes.]
- **Mitigation Plan:** [E.g., Apply Oversampling (e.g., SMOTE) or Undersampling techniques, and adjust class weights during training.]